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No experiment, no problem?

Causal Inference Methods in Quasi-Experimental Settings

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Hello!

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People Insights

Why are we here?

- Increasing appetite for data-driven insights to guide investments in human capital that go beyond correlations and reporting.
- We are scientists, but the applied world is hardly ever a perfect lab environment.
- There is no one-size-fits-all approach, but we can practice and discuss!

What are we doing?

- Introduce causal inference and the quasi-experimental design process.
- Apply concepts to a case study and determine appropriate quasi-experimental design.
- Laptops out! Test some hypotheses using statistical methods that mitigate the limitations of quasi-design.

What is Causal Inference & How Does It Work?

Causal inference is the statistical process of determining the independent effect of a treatment that is a component of a broader system.

- You can never observe the exact **same unit** (e.g., a specific person or business) **with and without a treatment at the same time**.
- You only get to see the **"factual" potential outcome**, while the alternative **"counterfactual" outcome** remains hidden.
- Can be viewed as a **missing data problem** where you must find ways to **impute the unobserved outcomes** and answer the **"what if"** intervention questions.

Let’s Go!

https://tinyurl.com/colab-workshop

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